Part 15 — Thirty more problems, worked: the second problem set

Three answers first, because two of your questions rest on a misunderstanding I caused by writing "6 of 54" without saying what the 54 were — and the count itself was wrong: the appendix holds 56.

Where the count comes from, and what it is not

56 is the number of problems in your own curriculum's appendix, labelled A1 to W3 (including B4 and G2, which an earlier extraction of mine missed) and starting at page 140 of `me-curriculum-wk01-66-and-56-problems.pdf`. It is not a number of datasets. I checked the whole document: **it supplies no data files at all** — no CSV, no accession to download, no spreadsheet. Every problem that needs numbers states them inline in its own text.

Of those 56, seven name a real paper and quote one measured parameter from it — the Fredens recoding count, the *Vibrio natriegens* generation time, the Newman amorphadiene partition coefficient, and so on. The other 47 use constructed numbers. That is what I meant, badly, by "6 of 54 use a real published dataset". See the corrigendum: the appendix holds 56.

Why not more? Two reasons, one of which is mine and one of which is not. I did not write those 56; they came with your curriculum, and I only verified their arithmetic — so their choice of constructed data was not a decision I made. And I cannot fix it by fetching real data now: this environment has no network access, so I can neither download a dataset nor check one. **If I wrote a problem claiming to use real measurements, I would be inventing them**, which is worse than a clearly labelled constructed example.

So the problems here come in two kinds, and both are honest

	Self-contained (X1–X24)	Download-required (Y1–Y6)
Data	Stated in the problem, or simulated with a known true answer	A real public dataset you download yourself
Advantage	You can check whether your method recovered the truth, which real data never lets you do	Real noise, real missing values, real file formats
Status here	**Every number was computed and is printed with its code**	Statement, source and code given; **I could not run them** — no network

The first kind is not a weaker substitute. A simulation with known truth is the only way to measure what a method *costs* — which is why X18 can tell you that a log-linear fit over the apparent exponential phase overstates μ by 97%, and no real growth curve could.

What the 24 new problems cover

Group	Problems	Skill
Mass, energy and redox balance	X1–X4	Yield ceilings with cofactors closed; route comparison; oxygen and heat duty
Kinetics and reaction engineering	X5–X8	Two-substrate fitting; substrate and product inhibition; thermal deactivation
Transport and reactor design	X9–X12	Dissolved CO_2 and pH; shear; gas holdup and flooding; heat removal
Experimental design and statistics	X13–X16	Two-way ANOVA with interaction; response surface; variance components; equivalence testing
Data analysis and instrument reading	X17–X19	Weighted calibration and carryover; growth-curve fitting; off-gas mass balance
Probability and library design	X20–X21	Coverage with uneven representation; barcode and UMI collisions
Economics and planning	X22–X24	Monte Carlo on selling price; campaign sizing under budget; vessel scheduling

They deliberately fill the gaps in the first 56: there was no ANOVA, no response-surface design, no variance-component analysis, no equivalence test, no heat balance, no shear or CO_2 calculation, no scheduling, and no Monte Carlo. Those are the eight things an examiner or a process group will ask about and the appendix did not cover.

Group 1 — Mass, energy and redox balance

X1 · extends weeks 18–19 · 1.5 h

Itaconic acid: a ceiling that carbon alone does not explain

Itaconic acid ($C_5H_6O_4$, 130.10 g/mol) is made from glucose through citrate: glucose → one pyruvate and one phosphoenolpyruvate; the PEP is carboxylated to oxaloacetate, the pyruvate is oxidised to acetyl-CoA losing a CO_2, the two condense to citrate, and cis-aconitate decarboxylase removes the second CO_2 to give itaconate.

1. Write the overall stoichiometry per mole of glucose and give the carbon and mass yield ceilings.
2. Count NADH, NADPH and ATP across the route. Is it redox-balanced?
3. Convert the surplus to an oxygen demand per gram of product.
4. Reported titers reach about 0.58, 0.72 and 0.86 of the ceiling in different hosts. What mass yields are those, and what would you measure to tell which constraint is binding?

Worked solution. One glucose gives one itaconate and loses one carbon as CO_2 at pyruvate dehydrogenase, while the other decarboxylation returns the carbon the PEP carboxylase fixed — so the net is **1 glucose → 1 itaconate + 1 CO_2**. Carbon yield 5/6 = **83.3%**; mass yield 130.10/180.156 = **0.722 g/g**. Glycolysis contributes 2 NADH and 2 net ATP, pyruvate dehydrogenase a third NADH, and neither the condensation nor the decarboxylation consumes reducing power, so the route leaves a **surplus of 3 NADH per product** and is not redox-balanced. Those electrons have to go to oxygen: 1.5 mol O_2 per mole of product, or **0.369 g O_2 per g of itaconate**. At 0.58, 0.72 and 0.86 of the ceiling the mass yields are **0.419, 0.520 and 0.621 g/g**.

Which constraint binds is answerable by measurement rather than argument: close a carbon balance including CO_2 and by-products — if carbon is unaccounted, it is leaving somewhere you have not looked; and measure the oxygen uptake rate against the 0.369 g/g the redox surplus demands — if the vessel cannot supply it, the ceiling is a transport ceiling and no pathway work will reach it.

What it teaches: a pathway can be carbon-efficient and redox-unbalanced, and the redox surplus converts directly into an oxygen duty that the reactor, not the strain, has to meet.

X2 · extends weeks 18–19 · 1.5 h

3-hydroxypropionate by two routes: which is actually better?

3-HP ($C_3H_6O_3$, 90.08) can be made from glycerol — dehydratase to 3-hydroxypropionaldehyde, then an aldehyde dehydrogenase — or from glucose through malonyl-CoA, where acetyl-CoA carboxylase fixes a CO_2 and two reductions give the product.

1. Give the carbon and mass yield ceiling of each route on its own substrate.
2. The malonyl-CoA route loses two CO_2 at pyruvate dehydrogenase. Why is its carbon ceiling still 100%?
3. Count ATP and NADPH per product for the malonyl-CoA route, and correct its ceiling for the carbon that supplying the NADPH costs.
4. At what glycerol-to-glucose price ratio do the two break even on feedstock cost alone?

Worked solution. The glycerol route is carbon-conservative — three carbons in, three out — giving 90.08/92.09 = **0.978 g/g**, and it generates one NADH rather than consuming reductant. The malonyl-CoA route loses two CO_2 making two acetyl-CoA, and then **acetyl-CoA carboxylase re-fixes one CO_2 per product**, so the six carbons of glucose end up in two three-carbon products: carbon yield **100%**, mass yield 2 × 90.08/180.156 = **1.000 g/g**. That is the headline number quoted for 3-HP, and it is real but conditional, because the route needs **1 ATP and 2 NADPH per product**. If the NADPH comes from a fully cycling oxidative pentose phosphate pathway at 12 NADPH per glucose, four NADPH costs a third of a glucose, so eight carbons are consumed for six in product and the **effective ceiling falls to 75%, about 0.75 g/g**.

Break-even on feedstock: cost per kg of product is price divided by yield, so the routes are equal when $P_{glycerol}/0.978 = P_{glucose}/0.75$, that is when glycerol costs **1.30 times** glucose per kilogram. Below that ratio the glycerol route wins on feedstock, which is why crude glycerol from biodiesel made this route attractive in the first place.

What it teaches: a 100% carbon yield can still be an overstatement once you charge the reductant to carbon, and the comparison between two routes is a price ratio, not a biology argument.

X3 · extends weeks 7 and 53 · 1.5 h

Isoprenoids: the mevalonate and MEP routes compared

Both routes make the five-carbon isopentenyl pyrophosphate. Mevalonate condenses three acetyl-CoA and loses one CO_2, spending 3 ATP and 2 NADPH. MEP condenses pyruvate and glyceraldehyde-3-phosphate, loses one CO_2, and spends a CTP, 3 NADPH and reduced ferredoxin.

1. Give the carbon ceiling to isoprene units for each route.
2. Amorphadiene is $C_{15}H_{24}$ (204.35). Compute its mass yield ceiling on glucose.
3. Tabulate ATP and reductant per amorphadiene for both routes.
4. Why do engineers import the mevalonate route into *E. coli*, which already has MEP?

Worked solution. Both routes convert six carbons into a five-carbon unit, so both have a carbon ceiling of **83.3%** — the common claim that MEP is intrinsically more carbon-efficient does not survive this count, because both lose exactly one CO_2 per C_5. Amorphadiene needs three C_5 units, so 15 carbons from 18, which is three glucose: **204.35/(3 × 180.156) = 0.378 g/g**. Per amorphadiene the mevalonate route spends **9 ATP and 6 NADPH**; MEP spends **3 CTP, 9 NADPH and 3 reduced ferredoxin** — cheaper in nucleoside triphosphate, dearer in reductant, and dependent on a ferredoxin pool that is tied to central metabolism in ways that are hard to push.

The reason for importing mevalonate is not thermodynamic but regulatory and practical: MEP in *E. coli* is feedback-controlled at its first step and embedded in native regulation, while a heterologous mevalonate operon is unregulated by the host and can be tuned by expression alone. That is the part 1 chapter 9 argument — **you import a pathway to escape regulation, not to improve stoichiometry**.

What it teaches: two routes with identical carbon ceilings can differ entirely in what they spend and in how controllable they are, and the engineering choice usually turns on the second.

X4 · extends weeks 18 and 44 · 2 h

Single-cell protein from methanol: the heat nobody budgets for

A 100 m³ fermenter grows a methylotroph on methanol to 30 gDW/L at $\mu = 0.15\ h^{-1}$, with a biomass yield of 0.40 g/g. Take biomass as $CH_{1.8}O_{0.5}N_{0.2}$ (24.6 g per C-mol, degree of reduction 4.2) and methanol as CH_4O (32.04 g/mol, degree of reduction 6).

1. From an electron balance, compute the oxygen demand per gram of biomass at yields of 0.35, 0.40 and 0.45 g/g.
2. Compute the biomass production rate and the oxygen uptake rate of the vessel.
3. Using 460 kJ of heat released per mole of oxygen consumed, compute the cooling duty.
4. With $U = 500\ W \cdot m^{-2} \cdot K^{-1}$ and a 10 K approach, what heat-transfer area is needed? A 100 m³ tank has roughly 110 m² of wall. What do you conclude?

Worked solution. The electron balance is $\gamma_S \cdot C_S = \gamma_X \cdot C_X + 4 \cdot O_2$. Per gram of methanol that gives $(6/32.04 - 4.2 \cdot Y/24.6)/4$ moles of O_2, so the oxygen demand per gram of biomass is **2.91, 2.38 and 1.96 g O_2 per g** at the three yields — methanol is a reduced substrate and costs far more oxygen than sugar does. At the design point the vessel makes $30 \times 0.15 \times 100$ = **450 kg of biomass per hour** and consumes 450×2.38 = **1,071 kg O_2 per hour**, which is 33.5 kmol/h. At 460 kJ/mol that is **4.28 MW of heat**, needing 368 m³/h of cooling water at a 10 K rise.

The area follows: 4.28 MW ÷ (500 × 10) = **855 m²**, against about 110 m² of jacket. The conclusion is structural, not a tuning problem — **a methanol fermentation at this density cannot be cooled through its wall** and needs internal coils or an external heat exchanger, which is why industrial methylotroph plants look the way they do. It is also why the same vessel on glucose, which demands roughly a third of the oxygen per gram, raises no such problem.

What it teaches: oxygen demand and heat release are the same calculation, and on a reduced substrate the heat, not the oxygen, is what sets the vessel design.

Group 2 — Kinetics and reaction engineering

X5 · extends week 3 and chapter S3 · 2 h

Fit a two-substrate enzyme, and see what a single-substrate fit hides

An NADPH-dependent reductase is assayed over a grid: six substrate concentrations from 0.05 to 1.6 mM at each of four cofactor concentrations from 0.05 to 2.0 mM, with 5% noise. The true parameters are V = 2.4, K_a = 0.30, K_b = 0.18 and K_{ia} = 0.45.

1. Fit the ternary-complex rate law and report parameters with standard errors.
2. Fit the ping-pong law to the same data and compare residual sums of squares.
3. Compute the apparent K_M for substrate A at the lowest and highest cofactor concentration.
4. What would a single-substrate fit at a saturating cofactor have reported, and why does that matter for a cofactor-swap campaign?

Worked solution. The ternary fit recovers **V = 2.43 ± 0.07, K_a = 0.347 ± 0.025, K_b = 0.193 ± 0.018, K_{ia} = 0.375 ± 0.073** — all within about one standard error of truth except K_{ia}, which is always the least identifiable parameter because it only shows up in the constant term. The ping-pong fit gives a residual sum of squares **5.9 times larger**, which with 24 points and one fewer parameter is a clear rejection; the diagnostic is the double-reciprocal pattern from chapter S3, intersecting against parallel.

The apparent K_M for A is $K_a + K_{ia}K_b/[B]$, which is **1.92 mM at [B] = 0.05** and **0.34 mM at [B] = 2.0** — a **5.6-fold difference in the same enzyme**. A single-substrate fit at saturating cofactor would report 0.34 and call it the K_M. For a cofactor swap that is the wrong number twice over: the quantity you are engineering is K_{ia}, the dissociation constant of the cofactor from the binary complex, which never appears in an apparent K_M; and a variant that looks improved at saturating cofactor can be worse at the intracellular concentration, which for NADPH sits far below saturation.

What it teaches: the K_M of a two-substrate enzyme is not a property of the enzyme, and the parameter a cofactor-swap campaign actually changes is invisible to the standard assay.

X6 · extends weeks 3 and 44 · 1 h

Substrate inhibition sets the feed set-point

An enzyme follows $v = V_{max}S/(K_M + S + S^2/K_{si})$ with K_M = 0.2 g/L and K_{si} = 8 g/L.

1. Derive the substrate concentration that maximises the rate and evaluate it.
2. What fraction of V_{max} is reachable at that optimum?
3. Tabulate the rate at 0.5, 1.3, 5, 20 and 50 g/L.
4. A fed-batch controller overshoots to 20 g/L. What has it cost, and how would you detect the overshoot without a substrate assay?

Worked solution. Differentiating and setting dv/dS = 0 gives **$S^* = \sqrt{(K_M K_{si})}$ = 1.26 g/L**, at which the rate is $V_{max}/(1 + 2\sqrt{(K_M/K_{si})})$ = **0.760 V_{max}** — note that the enzyme can never reach its own V_{max}, which is a feature of the rate law and not of the assay. The rate profile is **0.684, 0.760, 0.601, 0.285 and 0.138** of V_{max} at the five concentrations: the curve is asymmetric, falling slowly below the optimum and steeply above it.

An overshoot to 20 g/L runs at **37% of the achievable rate** — a 2.7-fold loss of productivity that looks exactly like a strain problem. Detecting it without a substrate assay is the useful part: the oxygen uptake rate falls as the metabolic rate falls, so the dissolved-oxygen trace *rises* while the feed is still running, and the respiratory quotient drifts. A rising DO during active feeding is the signature of inhibition or of a stalled culture, never of a healthy one.

What it teaches: with substrate inhibition the control objective is a set-point, not a maximum, and the cheapest instrument for finding the set-point is the dissolved-oxygen probe.

X7 · extends weeks 16 and 45 · 2 h

Product inhibition decides when to stop the batch

A fermentation follows $\mu = \mu_{max}(1 - P/K_P)\cdot S/(1 + S)$ with μ_{max} = 0.35 h^{-1} and K_P = 45 g/L, starting from 2 gDW/L and 120 g/L of substrate, with $Y_{P/S}$ = 0.45 g/g.

1. Integrate the system and tabulate biomass, product and substrate at 10, 20, 40, 60 and 80 h.
2. When does the culture reach 90% of the inhibition ceiling?
3. How much substrate is left at the end, and what does that mean for the yield on substrate fed?
4. What are the three engineering responses, and which does the model support?

Worked solution. The integration gives product at **34.9 g/L by 10 h, 44.7 by 20 h, and 45.0 thereafter** — the run is effectively over at 20 hours and the remaining sixty hours add nothing. It reaches 90% of the ceiling at **12.5 h**. Substrate stops at **70 g/L unconsumed** of the 120 fed, so the yield on substrate *fed* is 45/120 = **0.375 g/g** against a metabolic yield of 0.45 — the process throws away 58% of its feedstock, and the loss is invisible in a yield quoted on substrate consumed.

Three responses: harvest at 20 h and recycle the substrate, which fixes the feedstock loss and not the titer; remove the product in situ, which raises the ceiling (part 3 chapters 45 to 47); or engineer tolerance, which raises K_P. The model supports the first two immediately and says nothing about the third, because K_P is an input to it. The decisive measurement is whether the ceiling is tolerance or thermodynamic end-product inhibition: add product externally at the start and see whether growth stops at the same total.

What it teaches: a yield quoted on substrate consumed can hide a process that leaves most of its feed in the tank, and the harvest time is a design variable with a right answer.

X8 · extends weeks 3 and 22 · 1.5 h

Biocatalysis: the optimum temperature depends on the batch length

An enzyme's turnover has an activation energy of 50 kJ/mol and its thermal deactivation 250 kJ/mol, both referenced to 30 °C where k = 1.0 (relative) and k_d = 0.001 h^{-1}.

1. Tabulate the relative rate, the deactivation constant and the enzyme half-life from 25 to 60 °C.
2. Derive the total product formed in a batch of length t for a first-order decaying catalyst.
3. Find the temperature that maximises an 8-hour batch.
4. Repeat for 2, 24 and 72 hours. What is the general rule?

Worked solution. With activity decaying as $e^{-k_d t}$, the integrated product is proportional to $(k/k_d)(1 - e^{-k_d t})$. The table shows the trade sharply: at 40 °C the rate is 1.88 times the 30 °C value and the half-life is 29 h; at 50 °C the rate is 3.41 times but the half-life is **1.5 hours**; at 60 °C the enzyme is gone in minutes.

Batch length	Optimum temperature
2 h	47.4 °C
8 h	42.7 °C
24 h	39.1 °C
72 h	35.6 °C

The rule follows from the two activation energies: because deactivation is far more temperature-sensitive than catalysis, **the optimum falls as the batch lengthens**, by roughly 4 °C per tripling of duration here. A "optimum temperature" quoted from a ten-minute assay is therefore the wrong set-point for a day-long reaction — the assay optimum is near 50 °C and the 24-hour optimum is 39 °C. The same arithmetic is the argument for enzyme immobilisation, which usually raises k_d's activation energy and moves the whole curve.

What it teaches: an enzyme's optimum temperature is not a property of the enzyme but of the enzyme and the reaction time together.

Group 3 — Transport and reactor design

X9 · extends week 44 · 1.5 h

Dissolved CO_2: the variable nobody controls

CO_2 leaves the cells as fast as oxygen enters at RQ = 1, and it must be stripped by the same gas flow. Take Henry's constant for CO_2 as 0.025 M/atm at 37 °C and $pK_1 = 6.1$ for the CO_2/bicarbonate couple.

1. At CER values of 50, 150 and 250 mmol·L^{-1}·h^{-1} and k_La of 100 and 300 h^{-1}, compute the excess dissolved CO_2 at steady state.
2. Convert the worst case to an equivalent partial pressure.
3. Compute the bicarbonate in equilibrium with 1.5 mM dissolved CO_2 at pH 7.0.
4. Head pressure is often raised to 1.5 bar to help oxygen transfer. What does it do to CO_2, and what would you watch?

Worked solution. At steady state CER = $k_La(C - C^*)$, so the excess is CER/k_La. At 250 mmol·L^{-1}·h^{-1} and k_La = 100 h^{-1} that is **2.50 mM**, equivalent to **0.100 atm** of CO_2 — and at k_La 300 it falls to 0.83 mM. The bicarbonate in equilibrium with 1.5 mM dissolved CO_2 at pH 7.0 is $1.5 \times 10^{(7.0 - 6.1)}$ = **11.9 mM**, which is a substantial ionic load and consumes base in a pH-controlled vessel.

Raising the head pressure to 1.5 bar multiplies the dissolved CO_2 by 1.5 along with the oxygen, so the strain now sees **2.2 mM** where it saw 1.5. That is the trade the oxygen calculation usually ignores: pressure buys oxygen transfer and buys dissolved CO_2 at the same ratio. What to watch is base consumption and the pH control output, which rise before any growth effect appears, and — in a strain with a CO_2-sensitive carboxylation step — the by-product spectrum.

What it teaches: every measure that improves oxygen transfer also concentrates CO_2, and dissolved CO_2 is a scale-up variable that bench vessels never reveal.

X10 · extends week 44 · 1 h

Shear: what the impeller actually does to a cell

The smallest turbulent eddies have the Kolmogorov scale $\lambda = (\nu^3/\varepsilon)^{1/4}$, with $\nu = 10^{-6}$ m²/s and ε the dissipation rate per unit mass.

1. Compute λ at P/V of 100, 500, 1,000, 5,000 and 20,000 W/m³.
2. Compare with a 1 µm bacterium, a 15 µm animal cell and a 100 µm floc or pellet.
3. Local dissipation near the impeller is 20–50 times the mean. How does that change the picture?
4. What does this say about bacteria, animal cells and immobilised pellets?

Worked solution. The eddy scale is **56 µm at 100 W/m³, 31.6 µm at 1,000 and 15.0 µm at 20,000** — it falls only as the fourth root of power, so a 200-fold power increase shrinks the eddies by less than four times. Against that: a bacterium at 1 µm is smaller than the smallest eddy at any realistic power, so **bacteria are not damaged by shear in a stirred tank** and reports to the contrary are usually about the air–liquid interface or foam. An animal cell at 15 µm is comparable to the eddy scale at high power, which is why those cultures are run at a few tens of W/m³. A 100 µm floc or immobilisation pellet is larger than the eddies at every power in the table and **will be eroded**.

Near the impeller, dissipation is 20–50 times higher, dividing the scale by about 2.3 — so the 1,000 W/m³ case has 14 µm eddies locally, and the cells circulate through that zone repeatedly. The practical consequence is that pellet and floc processes are limited by impeller design and tip speed rather than by mean power, and that the standard advice to "reduce shear" for a bacterial culture is usually solving the wrong problem.

What it teaches: shear damage is a question of length scales, and for bacteria the answer is almost always that shear is not the problem.

X11 · extends week 44 · 1 h

Gas holdup and flooding: the limits of "add more air"

Take gas holdup as roughly $0.6 \cdot v_s^{0.7}$ with superficial velocity in m/s, and the flooding criterion for a Rushton turbine as a flow number $Q/(ND^3)$ above about 0.25, with N = 0.7 s^{-1} and D = 1.8 m.

1. Compute holdup at superficial velocities of 0.005, 0.02, 0.05 and 0.1 m/s, and the working volume lost.
2. Compute the flow number at gas rates of 0.05, 0.2, 0.5 and 1.0 m³/s and identify flooding.

3. Combine with X9: what happens to CO_2 stripping as you approach flooding?
4. State the three ways to raise oxygen transfer once flooding is reached.

Worked solution. Holdup rises from **1% at 0.005 m/s to 12% at 0.1 m/s** — the gas displaces broth, so a vessel pushed to high aeration loses more than a tenth of its working volume and its nominal titer per vessel falls with it. The flow numbers are **0.012, 0.049, 0.122 and 0.245**, so even 1 m^3/s is only just at the flooding threshold for this impeller at this speed; flooding is reached sooner at lower stirrer speed, which is exactly when a control system is trying to save power.

Beyond flooding the impeller no longer disperses the gas, bubbles coalesce and rise as a column, and k_La *stops rising with gas flow* — while CO_2 stripping, which depends on the same interfacial area, stops improving too. The three remaining levers are the ones from part 11 chapter 44: oxygen enrichment, which raises C* in proportion; head pressure, which does the same and worsens dissolved CO_2; and lowering demand by feeding more slowly.

What it teaches: aeration has a ceiling set by hydrodynamics, not by the compressor, and the holdup penalty means the last increment of oxygen transfer is paid for in working volume.

X12 · extends weeks 44 and 48 · 1.5 h

The sterilisation and cooling of a fed-batch, end to end

Using X4's methanol case as a contrast, take a 100 m^3 glucose fed-batch reaching 40 gDW/L with a specific oxygen uptake of 4 $mmol \cdot gDW^{-1} \cdot h^{-1}$, and 460 kJ released per mole of oxygen.

1. Compute the oxygen uptake rate, the required k_La at a 0.16 mM driving force, and the heat duty.
2. Compare the heat duty with X4's methanol case per unit volume.
3. How much cooling water at a 10 K rise, and what jacket area at U = 500 $W \cdot m^{-2} \cdot K^{-1}$?
4. At what cell density does the jacket alone stop being sufficient?

Worked solution. OUR = 40 × 4 = **160 $mmol \cdot L^{-1} \cdot h^{-1}$**, needing k_La = 160/0.16 = **1,000 h^{-1}** — already above what a large vessel delivers, which is the part 11 chapter 44 wall. The heat is 160 mmol/L/h × 100,000 L × 460 J/mmol = **2.04 MW**, against methanol's 4.28 MW in the same vessel: **methanol releases about twice the heat per unit volume** for a third less biomass, because it is the more reduced substrate.

Cooling water at a 10 K rise is 175 m^3/h, and the area required is 2.04 MW/(500 × 10) = **408 m^2** against roughly 110 m^2 of jacket. Setting the area equal to the jacket and solving backwards, the jacket alone suffices up to **about 11 gDW/L** at this specific uptake rate. That is the number worth remembering: **a high-cell-density process needs internal cooling from about 10 gDW/L upwards**, which is far below the densities everyone quotes as a target, and it is a capital decision taken at design time rather than a process adjustment.

What it teaches: the same oxygen number that sizes the sparger also sizes the cooling, and cooling runs out before oxygen does on a reduced substrate.

Group 4 — Experimental design and statistics

X13 · extends week 17 · 2 h

Two-way ANOVA: the interaction one-factor-at-a-time cannot see

Three media and two strains, four flasks each — 24 runs. Titers are generated with a true medium effect of 0 / 1.6 / 2.4 g/L, a strain effect of 1.1 g/L, an interaction of +0.9 g/L for strain 2 on medium 2, and a flask-to-flask standard deviation of 0.55 g/L.

1. Partition the total sum of squares into medium, strain, interaction and error, and build the ANOVA table with F and p.
2. What fraction of the variance does the model explain?
3. What would a one-factor-at-a-time campaign have concluded?
4. How many runs would the one-at-a-time route have used for the same information?

Source	SS	df	MS	F	p
Medium	22.09	2	11.04	50.25	<0.00001
Strain	6.73	1	6.73	30.64	0.00003
Interaction	3.01	2	1.51	**6.85**	**0.0061**
Error	3.96	18	0.220		

Worked solution. The model explains **89%** of the variance, and the interaction is significant at p = 0.0061 — the best medium is *not the same* for both strains. A one-factor-at-a-time campaign would have fixed one strain, found the best medium for it, then fixed that medium and compared strains, and would have reported both main effects correctly and **missed the 0.9 g/L interaction entirely**, because no run would have combined the second strain with the second medium at replication. It would then have recommended a strain–medium pairing that is not the best one available.

On run count, the one-at-a-time route needs 3 media × 4 + 2 strains × 4 = 20 runs to estimate the two main effects and still cannot estimate the interaction at all; the factorial spends 24 and estimates all three. That is the core design-of-experiments argument, and the honest caveat is that a factorial grows as the product of levels, so beyond three or four factors you move to a fractional design and accept confounding you choose deliberately.

What it teaches: interactions are the reason factorial designs exist, and one-at-a-time is not merely less efficient — it is blind to a whole class of effect.

X14 · extends weeks 17 and 41 · 2.5 h

Response surface: find an optimum in twelve runs and test the model

A central composite design in two factors — four factorial points, four axial points at ±1.414, and four centre replicates — on a true surface with an interior maximum and measurement noise of 0.25 g/L.

1. Fit the full quadratic model by least squares and report the coefficients.
2. Solve for the stationary point and classify it from the eigenvalues of the quadratic form.
3. Carry out a lack-of-fit test using the centre replicates.
4. Compare the information per run with a 5 × 5 grid.

Worked solution. The fit returns coefficients **9.15, +1.64 x_1, +1.29 x_2, −1.68 x_1^2, −1.28 x_2^2, −0.57 x_1x_2**. Solving $\nabla y = 0$ gives a stationary point at **x_1 = 0.417, x_2 = 0.411** with a predicted response of **9.75**, against the true optimum of 9.65 — recovered to within the noise from twelve runs. The eigenvalues of the quadratic form are **−3.66 and −2.26**, both negative, so the stationary point is a **maximum** and not a saddle; that check matters because a saddle point looks like an optimum in the coefficients alone and will send the next round in a useless direction.

The lack-of-fit test uses the four centre points to estimate pure error: **F = 0.53, p = 0.69**, so the quadratic model is adequate and there is no evidence of curvature it cannot capture. A 5 × 5 grid would need 25 runs to cover the same space, so the design delivers roughly **2.1 times the information per run** — and, unlike a grid, it gives a fitted surface, a stationary point and an adequacy test rather than a table of values.

What it teaches: the eigenvalue check and the lack-of-fit test are what separate a response-surface design from curve-fitting, and both come free with the centre replicates.

X15 · extends week 17 · 1.5 h

Variance components: replicate the right thing

Six strains, two flasks each, three HPLC injections per flask — 36 measurements. The true standard deviations are 0.80 g/L between strains, 0.35 between flasks and 0.12 between injections.

1. Estimate the three variance components from the nested mean squares.
2. Compute the standard error of a strain mean for the design as run.
3. Compare with three flasks × one injection and one flask × six injections, for the same 6 measurements per strain... and then for 3.
4. Where should the next unit of effort go?

Worked solution. The components come out at **analytical 0.014 (2%), flask 0.061 (8%), strain 0.730 (91%)** of the total variance. The standard error of a strain mean with two flasks and three injections is **0.181 g/L**. With three flasks and one injection each it is **0.158** — better, for half the chromatography; with one flask and six injections it is **0.252**, substantially worse despite the same number of numbers in the spreadsheet.

The rule the arithmetic enforces is that **injections are pseudo-replicates**: they estimate the analytical variance, which is 2% of the problem, and they do nothing about the flask variance that dominates what you can resolve between strains. The next unit of effort goes into more flasks, and the only reason to inject twice is to catch an instrument failure. The same logic recurs whenever a measurement is nested — wells within a plate, plates within a run, runs within a week — and the standard mistake is to treat the cheapest level as if it were replication.

What it teaches: replication at the wrong level buys precision you do not need; the variance components tell you which level to spend on.

X16 · extends week 17 · 1 h

Equivalence: proving two strains are the same

Strain A gives 12.1, 11.8, 12.4, 12.0, 11.9 g/L; strain B gives 12.3, 12.6, 11.9, 12.2, 12.4. You want to claim that a marker-free derivative is *equivalent* to the parent.

1. Run the conventional two-sample t-test and state what it does and does not establish.
2. Carry out two one-sided tests (TOST) against an equivalence margin of ±0.5 g/L, then ±1.0 g/L.
3. Report the 95% confidence interval for the difference and interpret it against both margins.
4. What margin would you pre-register, and on what basis?

Worked solution. The difference is **0.240 g/L** with **p = 0.16**, so the conventional test fails to reject — which establishes nothing, since absence of evidence at n = 5 is expected whatever the truth. TOST against ±0.5 g/L gives **p = 0.066, not equivalent**; against ±1.0 g/L it gives **p = 0.0006, equivalent**. The 95% interval is **−0.12 to +0.60 g/L**, which sits inside ±1.0 and not inside ±0.5 — exactly the same conclusion, read off the interval.

The margin has to be chosen before the data and on process grounds, not statistical ones: it is the largest difference that would not change a decision. For a marker-free derivative intended to replace the parent in production, a defensible margin is the run-to-run variability of the process itself, because a difference smaller than that is unobservable in manufacturing. Quoting "no significant difference" as equivalence is the error this problem exists to kill — and with these data it would have claimed equivalence at a margin the data cannot support.

What it teaches: failing to find a difference is not finding equivalence, and the equivalence margin is a process decision made before the experiment.

Group 5 — Data analysis and instrument reading

X17 · extends weeks 22 and 50 · 2 h

Calibration, weighting and the carryover that fakes a low-level peak

An HPLC calibration runs seven standards from 0.05 to 5.0 g/L with proportional noise — the standard deviation grows with concentration. The true slope is 57 area units per g/L with an intercept of 1.2.

1. Fit the calibration unweighted and with $1/\sigma^2$ weights; compare the slope, intercept and the back-calculated error on the three lowest standards.
2. Compute LOD and LOQ from a blank standard deviation of 0.05 area units.
3. At carryover of 0.1%, 0.5% and 2%, what does a 5 g/L sample leave in the following blank, as a fraction of the LOQ?
4. What injection order and what controls would you use?

Worked solution. Both fits recover the slope well — **57.26 unweighted, 57.46 weighted** — but the back-calculated error on the three lowest standards is **2.3% unweighted against 0.7% weighted**, because the unweighted fit lets the 5 g/L point dominate the line. LOD = 3.3σ/slope = **0.003 g/L** and LOQ = 10σ/slope = **0.009 g/L**.

The carryover result is the one worth internalising. A 5 g/L sample at 0.1% carryover leaves 0.005 g/L in the next injection, which is **57% of the LOQ**; at 0.5% it is **287%** and at 2% it is **1,149%** of the LOQ. In other words a quantifiable peak can be created entirely by the previous sample, which is how a trace product gets "detected" in a negative control. The defences are order and controls: inject from low to high concentration, put a blank after every high sample and *integrate it*, include a true negative control strain in the same sequence, and report carryover as a validated number rather than assuming it is negligible.

What it teaches: a detection limit is meaningless without a carryover figure, because carryover can exceed the LOQ by an order of magnitude.

X18 · extends weeks 10 and 17 · 1.5 h

Growth curves: fit the model, or misread the slope by a factor of two

Twenty-five hourly optical-density readings follow a Gompertz curve with a plateau of 2.2, μ_{max} of 0.42 h^{-1} and a lag of 3.0 h, with 0.03 of measurement noise.

1. Fit the Gompertz model and report all three parameters with standard errors.
2. Fit a straight line to ln(OD) over the window that looks exponential, 0.05 to 0.6, and compare the growth rate.
3. Explain the discrepancy.
4. Above what OD does Beer–Lambert fail, and what do you do about it?

Worked solution. The Gompertz fit recovers **A = 2.19 ± 0.01, μ_{max} = 0.420 ± 0.009 h^{-1}, lag = 2.98 ± 0.06 h** — all three parameters, to about 2%. The log-linear fit over the apparently exponential window returns **0.827 h^{-1}, which is 197% of the true value**.

The discrepancy is not noise and not a coding error: the window from OD 0.05 to 0.6 includes the *acceleration* out of lag, where the instantaneous slope of ln(OD) rises from zero toward μ_{max} and then falls again. Fitting a straight line through a sigmoid's inflection region gives the chord, not the tangent, and the chord here is nearly double. The fix is either to fit the full model, as in part 1, or to restrict the linear fit to a window verified to have constant slope by a rolling regression. On the instrument side, Beer–Lambert fails above about **OD 0.4** on a standard 1 cm path because of multiple scattering, so readings up to 2.2 must come from dilution or a short-path reader; using them raw flattens the curve and depresses the fitted plateau and rate together.

What it teaches: the "exponential phase" identified by eye is not exponential, and the most common growth-rate error in the literature is a factor of two from a linear fit over an inflection.

X19 · extends week 44 · 2 h

Off-gas analysis: OUR, CER and RQ from first principles

A 100 m^3 vessel is aerated at 0.5 vvm. Inlet air is 20.95% O_2, 0.04% CO_2 and the balance inert. Three outlet compositions are measured over a run.

1. Compute the inlet molar flow, then the outlet flow by an inert balance.

2. Compute OUR, CER and RQ for each case.
3. Compare with the naive calculation that assumes the outlet flow equals the inlet flow.
4. Convert the middle case to a specific uptake rate at 30 gDW/L and to the k_La it implies.

Outlet O_2 / CO_2	OUR (mmol·L⁻¹·h⁻¹)	CER	RQ	Naive OUR error
19.5% / 1.2%	20.4	15.5	0.76	−5%
18.2% / 2.8%	36.8	36.9	1.00	0%
16.8% / 4.5%	54.7	59.9	1.10	+2%

Worked solution. The inlet flow is 0.5 × 100,000 L/min × 60 ÷ 22.414 = **133,845 mol/h**. The outlet flow must be obtained from the inert balance, $F_{out} = F_{in} \cdot y_{inert,in} / (1 - y_{O_2} - y_{CO_2})$, because consuming oxygen and producing CO_2 at RQ ≠ 1 changes the total molar flow. The naive assumption is wrong by **−5% to +2%** here, which sounds small until you notice it is a systematic bias that varies through the run and so distorts the *shape* of the OUR trace, which is what you use to detect a feed problem.

The three RQ values read as a physiological narrative: **0.76** is growth on a substrate more reduced than the biomass with lipid synthesis under way, **1.00** is clean sugar oxidation, and **1.10** is the onset of overflow, with CO_2 appearing without a matching oxygen demand. The middle case at 30 gDW/L is **1.2 mmol·gDW⁻¹·h⁻¹** and implies a k_La of **230 h⁻¹** at a 0.16 mM driving force — comfortably achievable, which tells you this vessel is not oxygen-limited at this point and that the RQ excursion is physiological rather than a transfer artefact.

What it teaches: the off-gas analyser is the only instrument that reports metabolism continuously, and the inert balance is what makes its numbers trustworthy.

Group 6 — Probability and library design

X20 · extends weeks 28 and 30 · 1.5 h

Library coverage when the library is not uniform

A 10,000-member library is sampled at 30,000 and 50,000 clones. Representation is lognormal with coefficient of variation 0, 0.5, 1.0 and 2.0 — a realistic range for an oligo pool or a transformed library.

1. Derive the expected number of missing members for an arbitrary representation distribution.
2. Evaluate it at each CV and sampling depth.
3. By how much does the usual uniform-coverage rule understate the requirement?
4. How would you measure the CV of your own library, and what would you do about it?

CV of representation	Missing at 30,000 clones	at 50,000
0 (uniform)	498 (5.0%)	67 (0.7%)
0.5	979 (9.8%)	309 (3.1%)
1.0	1,843 (18.4%)	931 (9.3%)
2.0	3,389 (33.9%)	2,300 (23.0%)

Worked solution. For member *i* with probability p_i, the chance of never being sampled in k draws is e^{-kp_i}, so the expected number missing is Σe^{-kp_i} — which for a uniform library collapses to the familiar $Ne^{-k/N}$ and otherwise does not. At CV 1.0, **18.4% of the library is absent after 30,000 clones** where the uniform rule promises 5%; recovering the uniform figure needs about **2.6 times more clones**.

Measuring your own CV is a sequencing question: amplify the variable region from the pooled library and count reads per member, then take the coefficient of variation of the counts after correcting for the Poisson sampling of the sequencing itself. The remedies are upstream — normalise the oligo pool, use equimolar fragment assembly rather than a single ligation, and avoid growth steps between transformation and harvest, since a few hours of outgrowth imposes its own selection and widens the distribution further.

What it teaches: coverage rules assume uniformity, libraries are not uniform, and the correction is a factor of two to three in clones screened.

X21 · extends weeks 28 and 52 · 1 h

Barcodes and UMIs: when counting molecules stops working

A unique molecular identifier of length L has 4^L possible sequences. You tag n molecules.

1. Derive the expected number of collisions and the expected number of distinct identifiers recovered.
2. Evaluate for a 10-nt UMI at 10^5 and 10^6 molecules, a 12-nt at 10^6 and a 16-nt at 10^7.
3. By how much does a 10-nt UMI undercount 10^6 molecules?
4. What length would you specify for a barcoded strain library of 50,000 members, and why is the answer different from the UMI case?

Worked solution. Collisions follow the birthday calculation, $n(n-1)/2S$, while the distinct count recovered is $S(1 - e^{-n/S})$ — the saturation form. A 10-nt UMI (10^6 sequences) recovers **95.4% of 10^5 molecules but only 64.5% of 10^6**, so it **undercounts by 36%** at the higher load. A 12-nt UMI recovers 97.1% of 10^6, and a 16-nt recovers 99.9% of 10^7.

For a 50,000-member strain library the requirement is different and stricter: a UMI undercount is a quantitative bias you can correct for statistically, whereas two *strains* sharing a barcode is an identity error that silently merges two genotypes and cannot be corrected afterwards. You therefore design for essentially zero collisions rather than for acceptable saturation — with 50,000 members, a 12-nt barcode gives an expected 0.07 collisions and a 10-nt gives 1,250, so **12 nt is the minimum and 14 to 16 is the sensible specification**, with the additional requirement that barcodes differ at several positions so that a sequencing error cannot convert one into another.

What it teaches: the same birthday arithmetic has two different acceptance criteria depending on whether a collision biases a count or corrupts an identity.

Group 7 — Economics and planning

X22 · extends week 49 · 2 h

Monte Carlo on the selling price: which uncertainty actually matters

Take the part 4 chapter 49 model at a 50 g/L titer, 0.40 g/g yield, $0.40/kg sugar, $120M installed capital and 30,000 t/yr. Now treat five inputs as distributions: sugar price N(0.40, 0.06), yield N(0.40, 0.04), titer N(50, 8), capital lognormal with 25% spread, discount rate N(0.10, 0.02).

1. Compute the deterministic minimum selling price.
2. Run 20,000 Monte Carlo draws and report the median, P10 and P90.
3. Compute the probability of clearing $1.80, $2.00 and $2.50 per kg.
4. Rank the inputs by their correlation with the output and state where to spend the next engineering effort.

Worked solution. Deterministic MPSP is **$2.05/kg**. The Monte Carlo gives a median of **$2.08** with P10 **$1.76** and P90 **$2.47** — note that the median exceeds the deterministic value, because the lognormal capital distribution and the 1/titer term are both right-skewed, so a point estimate computed at the mean inputs is optimistic. Probabilities of clearing the three targets are **14%, 38% and 92%**.

Rank correlations with the output are **capital 0.66, sugar price 0.56, yield −0.38, titer −0.24, discount rate 0.23**, so the single biggest lever on the *uncertainty* is the capital estimate — not the yield, which is where a strain-engineering group would instinctively spend. That is the real use of the analysis: it separates what you can change from what dominates the risk, and it says that a firmer capital quote is worth more than a yield improvement of the size you could plausibly deliver. The complementary reading is that improving the yield shifts the whole distribution left, so it reduces the *level* while the capital estimate reduces the *spread*.

What it teaches: a point estimate at mean inputs is biased when the model is non-linear, and the input that dominates the risk is rarely the one the laboratory controls.

X23 · extends weeks 34 and 41 · 1 h

Sizing a campaign against a budget

A strain costs $120 to build and $35 to characterise. The budget for the round is $60,000.

1. How many strains does the budget buy?
2. For hit rates of 0.2%, 1% and 5%, how many strains give a 95% chance of at least one hit?
3. Which of those campaigns is adequately funded?
4. Halving the characterisation cost buys how many more strains, and is that the best use of the saving?

Worked solution. The budget buys **387 strains** at $155 each. A 95% chance of at least one hit needs ln(0.05)/ln(1 − p) strains: **1,496 at a 0.2% hit rate, 298 at 1%, and 58 at 5%**. So the campaign is adequately funded at 1% and 5% and **underfunded by a factor of four at 0.2%** — and 0.2% is a realistic hit rate for an unbiased library, which is precisely why the enrichment methods of part 8 chapter 29 matter: a thirteen-fold enrichment turns the hopeless case into the comfortable one.

Halving the characterisation cost raises the campaign from 387 to **436 strains**, a 13% gain, because build cost dominates. That is the answer to the fourth question: cheaper assays are the wrong place to optimise here, and the saving is better spent on anything that raises the hit rate, since the required number scales as 1/p while the affordable number scales only with the cost split. Before running the campaign at all, the number worth estimating is the hit rate itself — from a pilot of fifty clones, which costs $7,750 and bounds p well enough to decide.

What it teaches: screening campaigns fail on arithmetic more often than on biology, and the decision is between raising the hit rate and raising the throughput.

X24 · extends week 64 · 1.5 h

Scheduling two fermenters

Six runs are queued with durations of 7, 5, 7, 4, 6 and 5 days. Two identical vessels are available, and turnaround between runs is 1.5 days.

1. Compute the total vessel-days and the perfect-balance lower bound on the makespan.

2. Apply longest-processing-time first and give the resulting schedule.
3. Add turnaround and recompute.
4. What does this tell you about promising a date, and how does it connect to V1?

Worked solution. The runs total **34 vessel-days**, so the lower bound on two vessels is ⌈34/2⌉ = **17 days**. Longest-processing-time scheduling assigns A(7) + E(6) + D(4) = 17 days to one vessel and C(7) + B(5) + F(5) = 17 to the other — the heuristic hits the bound exactly, which it will not always do, and a quick check against the bound is how you know whether to look for a better schedule. Adding 1.5 days of turnaround between consecutive runs on the same vessel gives two changeovers per vessel and a makespan of **20 days**.

The connection to V1 is the one worth drawing: that 20 days is a *deterministic* figure built from most-likely run lengths, and V1 showed that such a plan has about a 1% chance of being met once durations are allowed to vary. A fermentation that overruns blocks its vessel, so delays propagate rather than absorbing — which means the right promise is the 80th percentile of a simulated schedule, and the right protection is a single visible buffer at the end rather than padding every run.

What it teaches: resource contention turns a list of durations into a scheduling problem with a computable lower bound, and the bound is what tells you whether to optimise the schedule or the plan.

Group 8 — Six problems on real public datasets

These are the "all from public datasets" problems you asked for, written as far as honesty allows. **I could not run them**: this container has no network access, so I cannot download the files, check their current format, or verify the answers. What follows is the source, the exact retrieval route, the questions, and code written against the documented file format — with my expectation of the answer stated as an expectation rather than a result. Treat the code as a starting point to debug, not as verified output.

	Dataset and source	What you do with it
Y1	**BiGG Models** — *E. coli* core model and iJO1366, as SBML and JSON	Reproduce the part 10 analyses on the real model: growth on glucose aerobically and anaerobically, the single-reaction deletion scan, flux variability at 99% of optimum, and pFBA. Compare with my hand-built model's 0.754 and 0.217 h^{-1} and account for every difference.
Y2	**Gene Expression Omnibus** — any bacterial RNA-seq count matrix with at least two conditions and three replicates	Normalise by median-of-ratios, test with a negative-binomial model, then repeat with the samples' labels shuffled. The shuffled run must return essentially nothing; if it does not, the pipeline leaks.
Y3	**BRENDA** — every k_{cat} and K_M reported for one enzyme of your capstone pathway	Tabulate by organism, temperature, pH and assay. Compute the spread of k_{cat} across reports, then propagate the worst case through the enzyme-constrained bound of chapter S5. Expect an order of magnitude.
Y4	**eQuilibrator** — $\Delta G'^{\circ}$ for every reaction of your pathway at pH 7.5, ionic strength 0.25 M	Compute the max-min driving force over the 1 µM–10 mM concentration window, identify the worst step, and test how much the answer moves when you take eQuilibrator's own uncertainty on each $\Delta G'^{\circ}$.
Y5	**ProteinGym** — a deep mutational scanning dataset for an enzyme	Score variants with a zero-shot predictor, compute the rank correlation with measured fitness, then compute the enrichment of true top-1% variants in your top 1,000 predictions. Compare with the 13-fold figure in part 8 chapter 29.
Y6	**Sequence Read Archive** — reads from an adaptive laboratory evolution experiment	Run breseq in polymorphism mode, read all three evidence tables including new junctions, and argue which one or two mutations are causal using the criteria of part 8 chapter 28.

Two notes on doing these well. **Record the accession, the version and the download date for every file** — sheet 9 of the lookup workbook has the fields, and a re-analysis six months later against a silently updated record is a real failure mode. And **run the self-contained problem first where one exists**: X22 before Y1's economics, X20 before Y5's enrichment, P1 and P2 before Y2. The simulated version tells you what the right answer looks like, so that when the real data give something different you can tell whether it is biology or a bug.

Closing notes to part 15

Answers to the four questions, in one place

Where does the count come from? It is the count of problems in your curriculum's appendix, A1 to W3, beginning at its page 140. Not datasets — the document supplies no data files at all.

What is the "datasheet" about? There isn't one. Each of the 56 states its own numbers inline, and seven of them quote one measured parameter from a named paper.

Why only those seven? I did not write the 56 — they came with your curriculum and I only verified their arithmetic, which all checked out. And I cannot retrofit real data now, because this environment has no network access; inventing data and calling it published would be worse than a labelled construction.

Can you make more? Yes — 24 new worked problems here, every number computed, plus 6 written against real public datasets with their retrieval routes, clearly marked as not run. That takes your set from 56 to **86**.

What the 24 add that the first 56 lacked

Gap in the original 56	Now covered by
No ANOVA, no interaction testing	X13
No response-surface or central composite design	X14
No variance components or nested replication	X15
No equivalence testing	X16
No heat balance or cooling duty	X4, X12
No dissolved CO_2, shear or gas holdup	X9, X10, X11
No off-gas mass balance	X19
No growth-curve model fitting	X18
No carryover or weighted calibration	X17
No Monte Carlo uncertainty propagation	X22
No scheduling or budget-constrained campaign sizing	X23, X24
No two-substrate kinetics or thermal deactivation	X5, X8

The six results most worth remembering

- **X4** — a methanol fermentation at 30 gDW/L needs 855 m² of heat-transfer area against about 110 m² of jacket. Cooling, not oxygen, sizes the vessel on a reduced substrate.
- **X8** — the optimum temperature falls from 47 °C for a 2-hour reaction to 36 °C for a 72-hour one. An assay optimum is the wrong set-point for a long batch.
- **X13** — a significant interaction ($p = 0.006$) that one-factor-at-a-time cannot detect at any sample size.
- **X15** — 91% of the variance is between strains and 2% is analytical, so extra injections buy almost nothing and extra flasks buy everything.
- **X18** — a log-linear fit over the apparent exponential phase overstates μ_{max} by 97%.
- **X22** — capital cost, not yield, dominates the uncertainty in the selling price.

On the numbers in this part

Every figure in X1 to X24 was computed in this session, and the parameters are printed in each problem. The ones that carry assumptions worth challenging: the 460 kJ per mole of oxygen in X4 and X12, which sets both heat duties; the holdup correlation in X11, which is a rough empirical form and varies with medium and sparger; the 20–50× local dissipation factor in X10; and the input distributions in X22, which I chose. Where a problem simulates data, the true parameters are stated so you can check whether your method recovered them — which is the whole point of the self-contained kind.

If you want more after these, the categories with the most room left are downstream separations (chapter S1 has the theory and only two problems), quantitative proteomics and metabolomics (chapter S5, no problems yet), and consortia

dynamics (chapter S4, no problems yet). Say which and I will work them the same way.